



Kumar Utkarsh

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B.Tech - Data Science and Artificial Intelligence

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. Major	Indian Institute of Technology, Guwahati	7.94 (Current)	2025-Present
Senior Secondary	CBSE Board	94.5%	2024
Secondary	CBSE Board	97.8%	2022

PROJECTS

- Gas Optimizer** Feb 2026 - March 2026
Kriti 2026, Inter Hostel Technical Competition, IIT Guwahati, Web3Assam Github-Project Link
 - Built a gasless transaction platform on Ethereum enabling users to execute batched smart contract calls without holding ETH, using EIP-712 typed signatures, EIP-2771 trusted forwarding, and a Fastify relayer service.
 - Developed a React + Vite + wagmi frontend with real-time relayer health monitoring, wallet-based auth, and a dashboard for batch submission and gas analytics deployed on Vercel with a Railway-hosted relayer.
 - Achieved ~45 percent gas savings on multi-call batches vs sequential transactions, validated through Foundry unit, fuzz, invariant, and integration tests across 7 core Solidity contracts on Sepolia testnet.
- BDH Interpretability explorer** Jan 2026 - Feb 2026
Kriti 2026, Inter Hostel Technical Competition, IIT Guwahati, Pathway Github
 - Built a 9-page interactive visualization platform using React, TypeScript, Three.js, and FastAPI for exploring a biologically-inspired post-transformer language model.
 - Trained multilingual BDH models on Europarl corpus with PyTorch (bfloat16, A100) and merged via neuron concatenation into a 24,576-neuron polyglot model.
 - Implemented monosemanticity probing, Hebbian synapse tracking, 3D graph topology, and sparsity analysis with a 6-phase extraction pipeline deployed on Hugging Face Spaces.
- Netflix-yield strategy** October 2025 - November 2025
Hackathon Project, Octant - v2 DeFi Hackathon Github
 - Identified the inefficiency of recurring subscription payments and built a DeFi-powered solution that deposits 3,600 USDC once into an ERC-4626 vault, channels the ~5 percent APY yield from Aave V3 to autonomously pay monthly Netflix subscriptions, keeping the principal fully protected.
 - Engineered end-to-end Solidity smart contracts implementing the Octant V2 donation-share pattern, with a 30-day time-lock, and graceful revert logic for insufficient yield, achieving test coverage via Foundry.
 - Integrated MetaMask Card and built a React + Wagmi v2 + RainbowKit frontend with live blockchain data polling, enabling a fully automated payment pipeline with no manual intervention.

TECHNICAL SKILLS

Programming: C, C++, Solidity, Python

Version Control: Git, GitHub

Finance: Blockchain, Fundamental Analysis

Web Technologies: HTML, CSS, ReactJS, Vercel, php

Database: MySQL, IPFS

Operating Systems: Windows, Linux

KEY COURSES TAKEN

- Computer Science:** Data Structures and Algorithms (Theory & Lab), Data Science (Theory & Lab), Discrete Mathematics
- Mathematics:** Probability Theory and Random Processes, Elementary Number Theory, Linear Algebra, Real analysis and vector calculus

POSITIONS OF RESPONSIBILITY

- Defi and Blockchain Head**
Finance and Economics Club, IITG Apr 2026 - present
 - Leading cross-functional teams of **25+ developers** and executing strategic partnerships within the Web3 Ecosystem.

ACHIEVEMENTS

- 1344 Rated** ,in Codeforces platform, profile - Kumar_Utkarsh9263 2025
- Octant v2 DeFi Hackathon**, Got the most Liked project and got their POAP token for free as reward. 2025
- 3 Star(1613 rating)**, Codechef platform ranked 23 in Starters 234 contest, username - neat_glim_65 2025
- World Quant Brain Challenge**, Secured Team Rank of 2000 among 80,000+ teams 2025